Exercise 1 — Lab: Learning the Concepts with a Drone and a Robot Arm

Optimization-Based Control Methods · Linear Systems and Linear Controllers

The Concepts file explains the theory in symbols. This lab shows the same ideas on two physical systems, using Python plots and animations. Every term from Exercise 1 appears here as something you can **see**:

Concept	Where you see it
eigenvalue, mode, λ^k	Experiment 1
stability, unit circle, repeated eigenvalues	Experiment 2
controllability, W_c, reachable subspace, PBH, minimum steps	Experiment 3
observability, W_o, unobservable subspace, PBH	Experiment 4
state feedback $u = -Kx$, closed loop $A - BK$, pole placement	Experiment 5
Task 1: tracking error $(I - A + BK)^{-1}(A - I)x_{\text{ref}}$	Experiment 6

How to run it

```
cd Exercise-01_Linear-Systems-and-Controllers/lab
pip install numpy scipy matplotlib pillow
python run_all.py                 # runs all 6 experiments, writes figures/ and GIFs
python exp3_controllability.py    # or run a single experiment
python exp5_state_feedback.py --show   # --show opens interactive plot windows
```

Each script prints its numbers (matrices, ranks, eigenvalues) to the terminal and saves its figures into `lab/figures/`. The best way to learn from it is to **change a number in a script and re-run it**. Every experiment ends with "Try it yourself" suggestions.

The two systems

System 1 — a planar quadcopter (drone)

Imagine a drone flying in a vertical plane, seen from the side. It has two rotors. Their **total thrust** lifts it, and the **difference** between them tilts it.

State (6 numbers that describe the drone completely at one moment):

$$x = [y \quad z \quad \theta \quad v_y \quad v_z \quad \omega]^\top$$

	meaning
y	horizontal position [m]
z	altitude, measured from the hover height [m]
θ	tilt (pitch) angle [rad]
v_y, v_z	horizontal and vertical velocity [m/s]
ω	tilt rate [rad/s]

Input (2 numbers the controller commands): $u = [\,\delta F,\ \tau\,]^\top$. Here δF is the extra thrust above what's needed to hover, and τ is the torque from the difference between the two rotor thrusts.

Physics, linearised about hover. The full equations contain $\sin\theta$ and $\cos\theta$. For small tilts, $\sin\theta \approx \theta$ and $\cos\theta \approx 1$, and gravity is exactly cancelled by the hover thrust. With mass $m = 0.5$ kg, inertia $J = 0.004$ kg m² and a small air drag $c = 0.2$ 1/s:

$$\ddot{y} = -g\,\theta - c\,\dot{y} \qquad \text{(tilting the thrust vector pushes sideways)}$$

$$\ddot{z} = \frac{\delta F}{m} - c\,\dot{z} \qquad \text{(extra thrust lifts)}$$

$$\ddot{\theta} = \frac{\tau}{J} \qquad \text{(torque spins the drone)}$$

These are linear, so they can be written as $\dot{x} = A_c x + B_c u$ (continuous time).

From continuous to discrete time. A digital controller measures and acts every $T_s = 0.1$ s and holds the input constant in between (a *zero-order hold*). The exact discrete model is then

$$x_{k+1} = Ax_k + Bu_k, \qquad A = e^{A_c T_s}, \quad B = \int_0^{T_s} e^{A_c s}\,ds\; B_c,$$

which is the model class of Exercise 1. The function c2d in `lab_common.py` computes this. A useful rule that follows from it:

$$\boxed{\lambda_{\text{discrete}} = e^{\lambda_{\text{continuous}}\,T_s}}$$

So the continuous-time stability condition ($\operatorname{Re}\lambda_c < 0$) becomes the discrete-time one ($|\lambda| < 1$), and a pure integrator ($\lambda_c = 0$) becomes $\lambda = e^0 = 1$.

System 2 — a robot arm joint

This is a single link of length $l = 0.5$ m and mass $m = 1$ kg, hanging under gravity and driven by a motor torque τ. It is linearised about the hanging-down position, with $T_s = 0.05$ s:

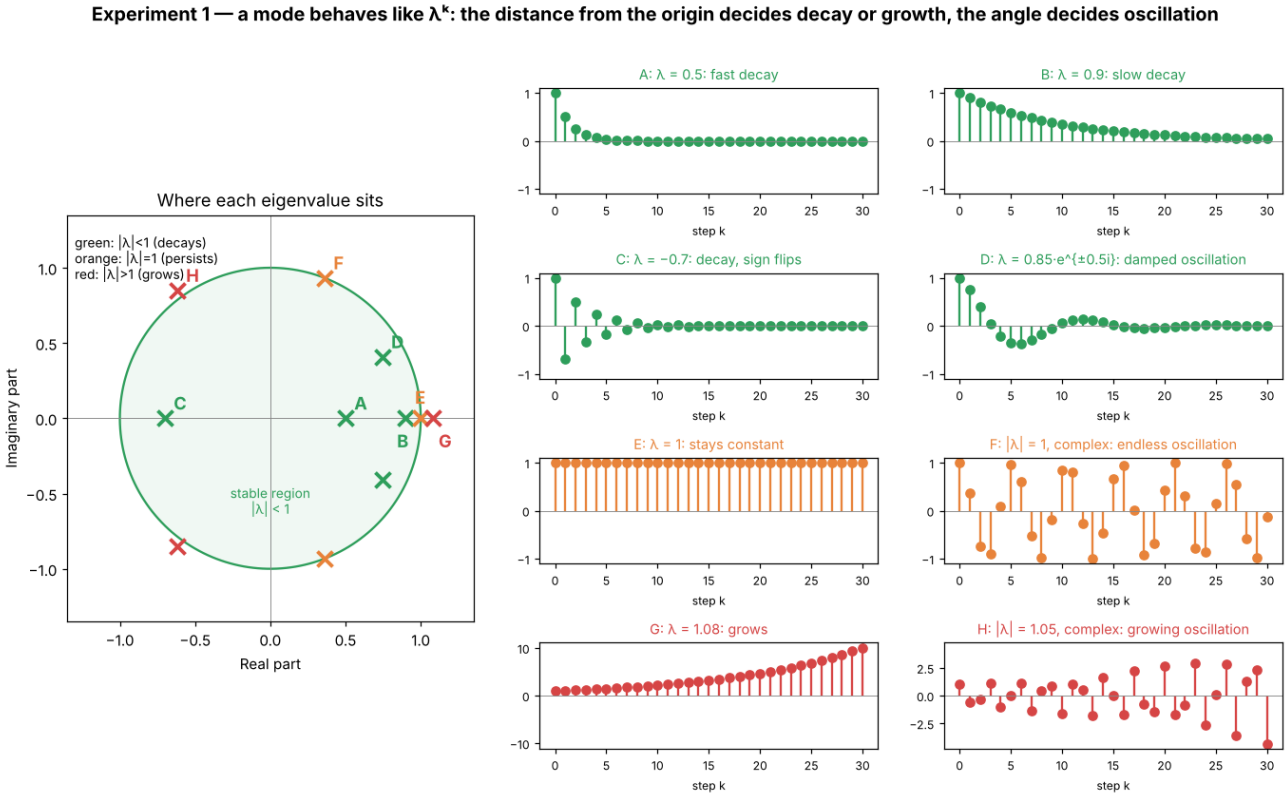
$$\ddot{\theta} = -\frac{g}{l}\theta - \frac{b}{ml^2}\dot{\theta} + \frac{\tau}{ml^2}, \qquad x = [\theta,\ \omega]^\top, \quad u = \tau.$$

The important physical fact for Task 1: **gravity always pulls the arm back down**. Holding it at any angle other than 0 requires a constant torque.

Experiment 1 — What an eigenvalue does

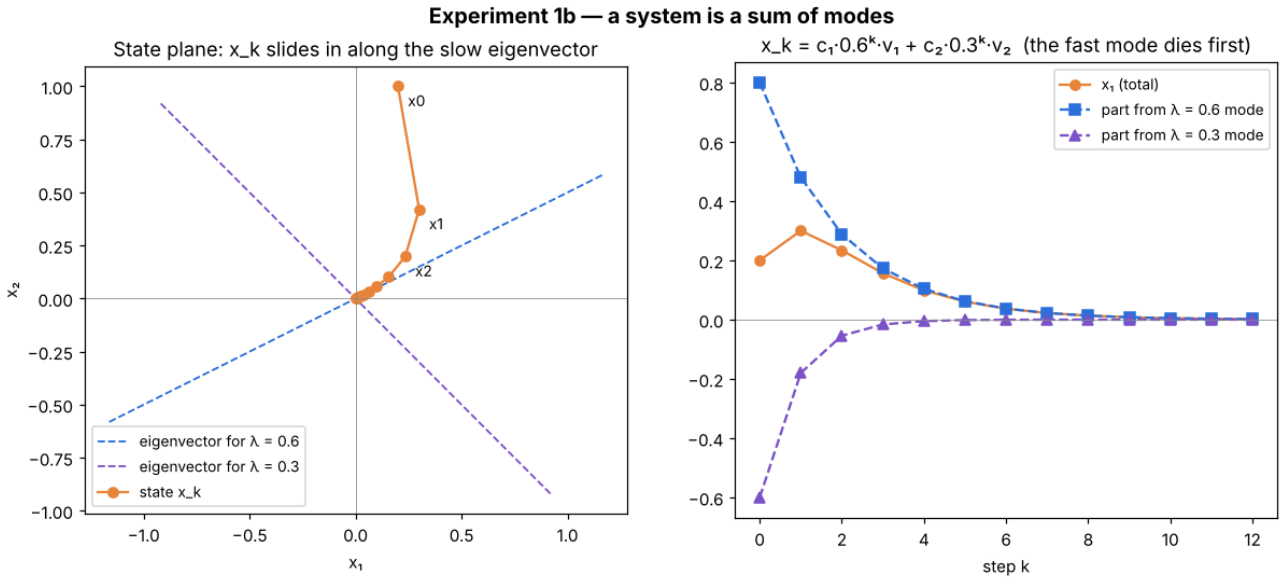
Script: lab/exp1_eigenvalues_and_modes.py · **Concepts:** Concepts §2, §3.3

Along an eigenvector, the system $x_{k+1} = Ax_k$ simply multiplies the state by λ every step, so $x_k = \lambda^k x_0$. This experiment places eight eigenvalues around the unit circle and plots λ^k for each.



What to notice

- **Distance from the origin, $|\lambda|$, decides growth.** Inside the circle (A, B, C, D) the sequence dies out; on the circle (E, F) it keeps the same size forever; outside (G, H) it grows. After 30 steps: $0.5^{30} \approx 10^{-9}$, $0.9^{30} \approx 0.04$, $1^{30} = 1$, $1.08^{30} \approx 10$.
- **Closer to the origin = faster.** Compare A ($\lambda = 0.5$, gone after about 5 steps) with B ($\lambda = 0.9$, still visible after 30 steps).
- **The angle decides oscillation.** A negative real λ (C) flips sign every step. A complex pair (D, F, H) rotates, which shows up as oscillation. The larger the angle, the faster the oscillation.



A whole system is a sum of modes. For the matrix of practice problem 5 (eigenvalues 0.6 and 0.3), write the initial state as $x_0 = c_1 v_1 + c_2 v_2$. Then

$$x_k = c_1\, 0.6^k\, v_1 + c_2\, 0.3^k\, v_2.$$

On the right, the purple ($\lambda = 0.3$) part dies first. On the left, after a couple of steps the state slides into the origin **along the slow eigenvector**, because only the slowest mode is left. In general, **the eigenvalue with the largest $|\lambda|$ sets how fast the whole system settles.**

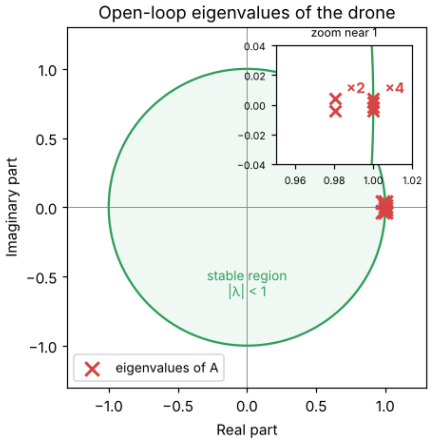
Try it yourself: in the script, change case B to $\lambda = 0.99$ or $\lambda = -0.99$. How many steps until it is below 0.1? (Answer: $k = \ln 0.1 / \ln 0.99 \approx 229$.)

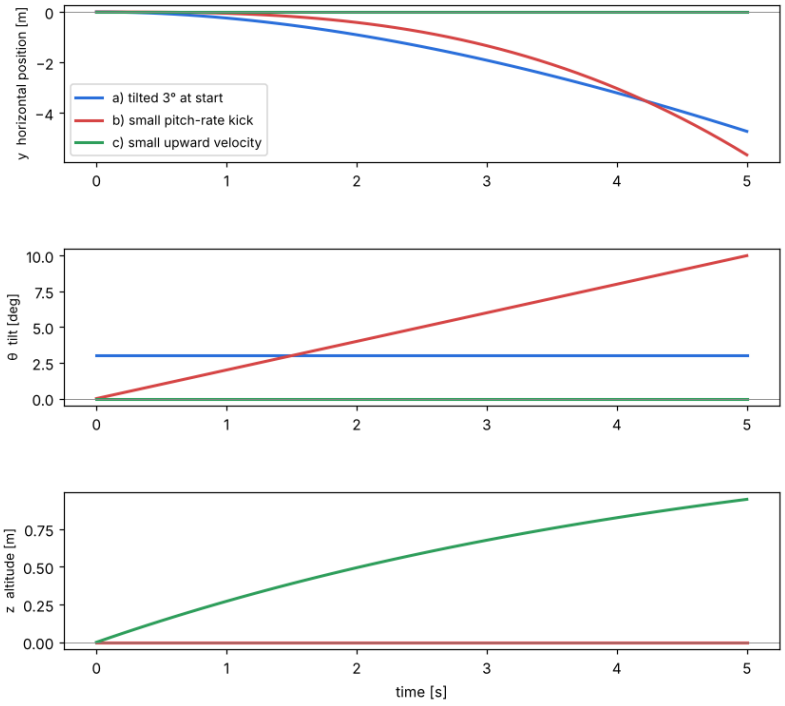
Experiment 2 — The drone without a controller

Script: `lab/exp2_drone_open_loop.py` · **Concepts:** Concepts §2.6, §3.3, §4.1

Fly the drone with **zero input** (exact hover thrust, no torque) from three slightly disturbed starts.

Experiment 2 — without feedback, tiny disturbances never die out: the drone drifts away






```
What the eigenvalues say
• 4 × λ = 1  (y, z, θ, ω):
    integrators, e^(0·Ts) = 1
• 2 × λ = e^(−c·Ts) = 0.980
    velocities, damped by drag
• λ = 1: 4 repeats, only 2 eigenvectors
    → growth like k, k², k³
• Not asymptotically stable:
    hovering needs a controller
```

The eigenvalues (printed by the script):

```
Continuous-time eigenvalues  : [-0.2 -0.2  0.   0.   0.   0. ]
Discrete-time eigenvalues    : [0.9802 0.9802 1.     1.     1.     1.    ]
Check lambda_d = e^{lambda_c Ts}: [0.9802 0.9802 1.     1.     1.     1.    ]
lambda = 1: algebraic multiplicity 4, geometric multiplicity 2
```

- **Four eigenvalues at exactly 1.** Position y, altitude z, tilt θ and tilt rate ω are all *integrators*: nothing pulls them back. Continuous $\lambda_c = 0$ becomes $e^{0 \cdot T_s} = 1$.
- **Two eigenvalues at $e^{-cT_s} = 0.980$.** Air drag slowly damps the two velocities.
- **$\lambda = 1$ repeats 4 times but has only 2 eigenvectors.** When an eigenvalue has fewer eigenvectors than repeats, the matrix is not diagonalisable (it has a *Jordan block*), and the state can grow like k, k^2, k^3 even though $|\lambda| = 1$. This is why the stability condition says a repeated eigenvalue on the unit circle needs enough eigenvectors (Concepts §4.1).

What the three flights show

start	what happens	why
a) tilted 3°	tilt stays 3°, the drone accelerates sideways: $y = -4.7$ m after 5 s	tilt is an integrator (stays), and it keeps pushing v_y
b) small tilt-rate kick	tilt grows linearly to 10°, y runs away even faster (-5.7 m)	the chain $\omega \to \theta \to v_y \to y$: a constant ω makes $\theta \sim k$ and $y \sim k^3$
c) small upward velocity	rises and *stays* at the new height (+0.95 m)	drag stops v_z ($\lambda = 0.98$), but nothing brings z back ($\lambda = 1$)

Animation "Drone drifting": open `lab/figures/exp2_drone_drift.gif` in the repository.

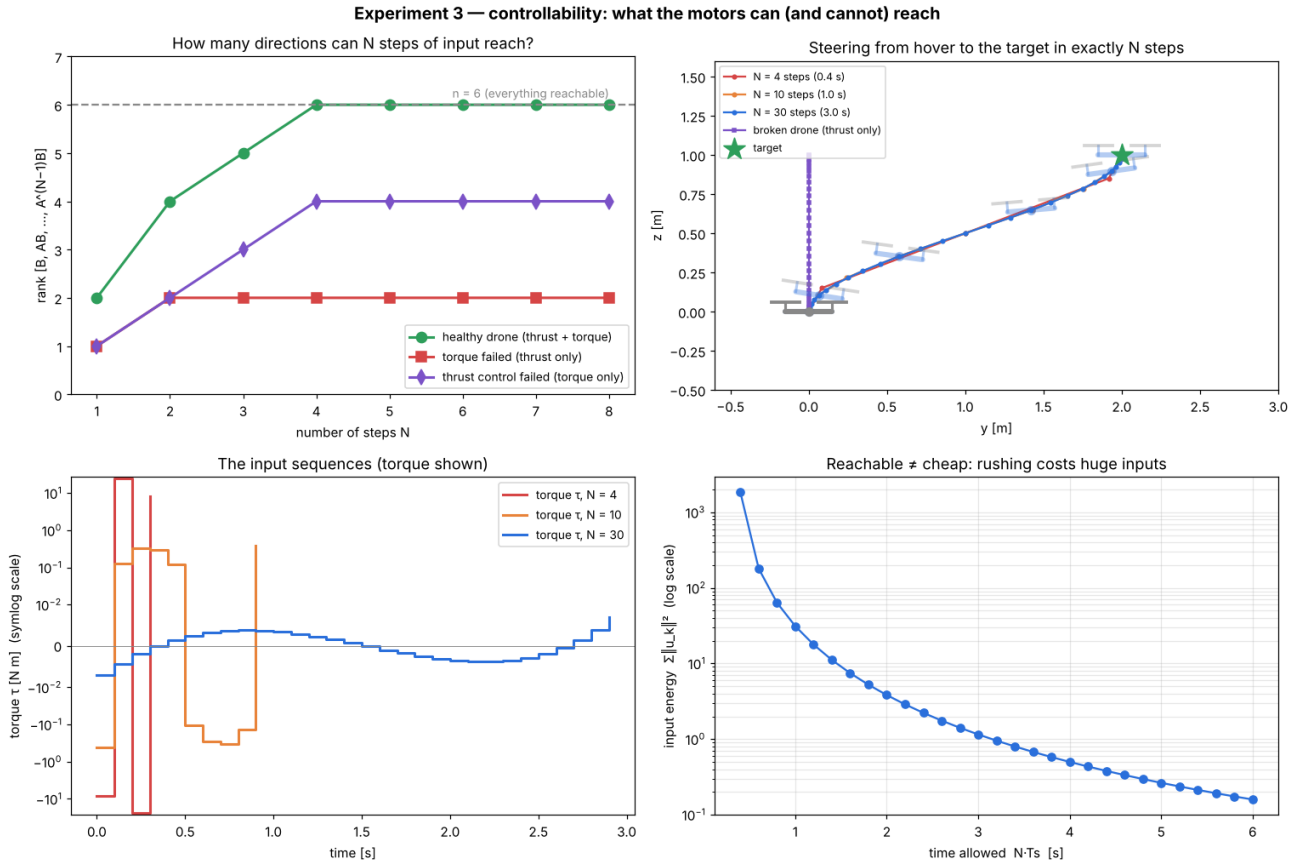
Conclusion: not all $|\lambda| < 1$, so the drone is **not asymptotically stable**. A tiny disturbance never dies out, and some grow without bound. A real drone cannot hover without feedback.

Try it yourself: set `drag=0` in `planar_quadcopter()` and re-run. All six eigenvalues become exactly 1. Which flight changes the most?

Experiment 3 — Controllability: what can the motors reach?

Script: `lab/exp3_controllability.py` · **Concepts:** Concepts §5 · Solutions Task 0.1–0.2

We compare three drones: a **healthy** one; one whose **torque control failed** (only total thrust δF works); and one whose **thrust control failed** (only torque τ works). Their input matrices are B, the first column of B, and the second column of B.



Top-left: rank of $W_c(N) = [B,\ AB,\ \ldots,\ A^{N-1}B]$ as the number of steps N grows. The rank is the number of independent directions that N steps of input can push the state in.

```
healthy drone (thrust + torque):     rank W_c = 6 of 6   -> reachable
torque failed (thrust only):         rank W_c = 2 of 6   reachable subspace: z, vz
thrust control failed (torque only): rank W_c = 4 of 6   reachable subspace: y, theta, vy, o
```

- **Healthy:** the rank climbs $2 \to 4 \to 5 \to 6$ and is full after 4 steps. The system is **reachable**, so every state can be reached.
- The rank **stops growing** once it stalls. That is Cayley–Hamilton (Concepts §5.3): $n = 6$ steps are always enough, and more steps never add new directions.
- **Thrust only:** stuck at rank 2. Thrust can only change altitude, never sideways position or tilt. The **reachable subspace** is $\{z, v_z\}$.

- **Torque only:** stuck at rank 4, the sideways chain $\tau \to \omega \to \theta \to v_y \to y$. Altitude is unreachable.

The **PBH test** confirms which eigenvalues are uncontrollable. For example, for the thrust-only drone:

```
PBH  lambda = 0.9802: rank[lambda I - A, B] = 5  -> UNCONTROLLABLE mode
PBH  lambda = 1.0000: rank[lambda I - A, B] = 5  -> UNCONTROLLABLE mode
```

The uncontrollable eigenvalues include $\lambda = 1$, which does not decay by itself. So the broken drones are **not controllable and not even stabilizable**: no controller K can stabilise their sideways motion. Compare Task 0, where $\lambda = 1$ was the problem as well.

Top-right: steering from hover to a target in exactly N steps. Solving $W_c(N)\,\bar{u} = x_f - A^N x_0$ gives the input sequence. Section 5.4 of the Concepts does this by hand for 3 states; here the script uses the pseudo-inverse, which picks the *minimum-energy* solution. All three healthy runs reach the target ★ exactly ($y = 2$, $z = 1$, level, at rest). The broken drone (purple) goes straight up and stops at $y = 0$, because it can only reach the part of the target that lies in its reachable subspace.

Bottom: reachable does not mean cheap.

N (time)	max torque	input energy $\sum\|u_k\|^2$
4 steps (0.4 s)	24.5 N m	1830
10 steps (1.0 s)	0.40 N m	31
30 steps (3.0 s)	0.007 N m	1.2

The minimum number of steps is 4, the first N with full rank. But reaching the target that fast needs a torque thousands of times larger than taking 3 seconds. A real motor would saturate. Controllability tells you *whether* a state can be reached; how *costly* it is to reach is a separate question. Optimisation-based control, the topic of this course, is about exactly that trade-off.

Animation "Steering animation": open `lab/figures/exp3_steering.gif` in the repository.

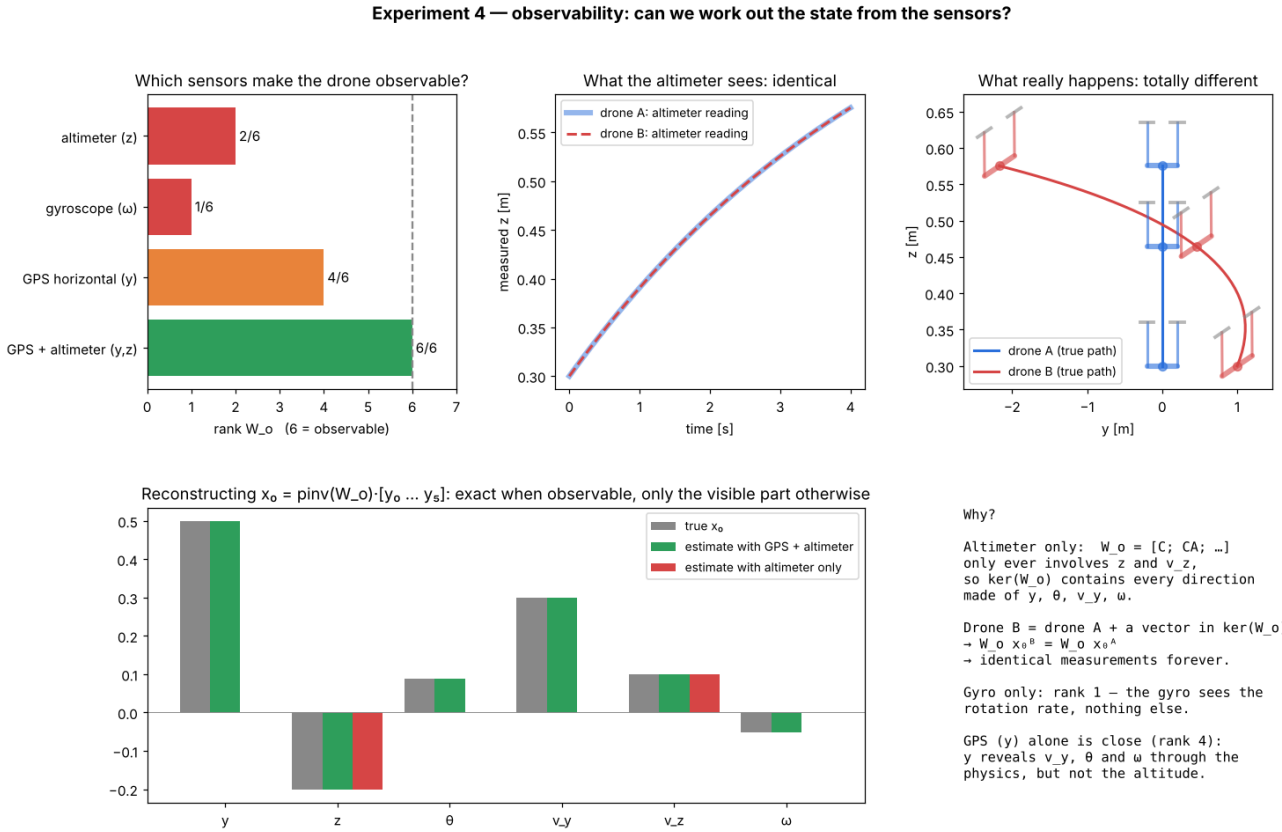
Try it yourself: set `xf = np.array([0, 1.0, 0, 0, 0, 0])` (straight up) and re-run. Now the thrust-only drone reaches it too. Why? (Because $[0, 1, 0, 0, 0, 0]$ lies in its reachable subspace $\{z, v_z\}$.)

Experiment 4 — Observability: what can the sensors tell us?

Script: lab/exp4_observability.py · **Concepts:** Concepts §6 · Solutions Task 0.3–0.4

A controller $u = -Kx$ needs the full state, but real drones only have sensors. Each sensor set is a C matrix, $y_k = Cx_k$:

```
altimeter (z)            rank W_o = 2 of 6   unobservable states: y, theta, vy, omega
gyroscope (omega)        rank W_o = 1 of 6   unobservable states: y, z, theta, vy, vz
GPS horizontal (y)       rank W_o = 4 of 6   unobservable states: z, vz
GPS + altimeter (y,z)    rank W_o = 6 of 6   unobservable states: none   -> observable
```



Top: two drones, identical measurements. Drone A starts 0.3 m high. Drone B starts at the same place **plus** a vector $h = [1,\ 0,\ 4°,\ 0.4,\ 0,\ 0]$ that lies in $\ker(W_o)$ for the altimeter, so $W_o h = 0$. By linearity, $W_o x_0^B = W_o x_0^A + W_o h = W_o x_0^A$. The altimeter readings are therefore **exactly identical** (the difference printed by the script is 0.00). Yet drone A only moves up, while drone B ends up more than 3 m from where it started, at $y = -2.2$ m. The altimeter has no way to tell them apart. That is what an **unobservable subspace** means: the directions that produce zero output.

Bottom: reconstructing x_0 from the first 6 measurements. Solving $W_o x_0 = [y_0; \ldots; y_5]$ with the pseudo-inverse gives:

```
true x0                    = [ 0.5   -0.2   0.0873  0.3   0.1  -0.0524]
estimate (GPS + altimeter) = [ 0.5   -0.2   0.0873  0.3   0.1  -0.0524]   exact
estimate (altimeter only)  = [ 0.    -0.2   0.      0.    0.1   0.    ]   only z, vz
```

With an observable sensor set the reconstruction is exact. With the altimeter alone, only the **observable component** (z, v_z) is recovered, and the unobservable directions come back as 0. This is the same split as Task 0.4: $x_0 = x_{\text{obs}} + x_{\text{unobs}}$, and the outputs only reveal x_{obs}.

Why GPS alone almost works (rank 4). It measures only y, yet the physics links the other sideways states to it. y changes because of v_y, v_y changes because of θ, and θ changes because of ω. Watching how y evolves over several steps reveals all four. The rows $CA, CA^2, \ldots$ of W_o are exactly "how the measurement evolves". Altitude never affects y, so it stays hidden.

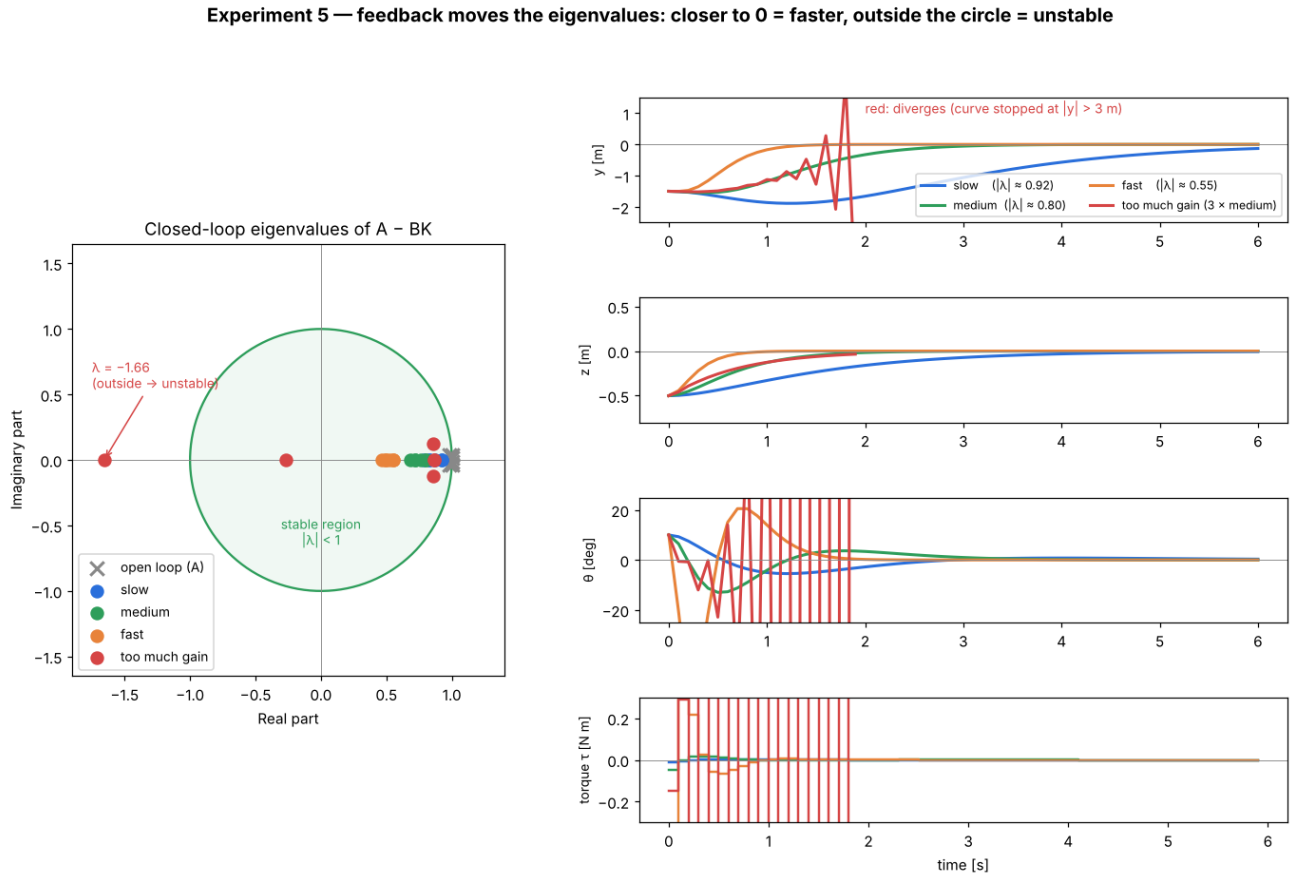
Animation "Same sensor, different reality": open `lab/figures/exp4_same_sensor_different_reality.gif` in the repository.

Try it yourself: add a sensor that measures the tilt, `sel(0, 1, 2)`. Does the rank change? (No, it is already 6. Extra sensors do not add rank once you have full rank, but they help against noise.)

Experiment 5 — State feedback: moving the eigenvalues

Script: lab/exp5_state_feedback.py · **Concepts:** Concepts §5.7, §7.1

With $u_k = -Kx_k$ the drone becomes $x_{k+1} = (A - BK)x_k$. Because the drone is **reachable**, K can place the eigenvalues of $A - BK$ anywhere (pole placement). We design three controllers whose eigenvalues all lie at roughly 0.92, 0.80 and 0.55, and a fourth controller that is simply the medium one multiplied by 3. All four start from 1.5 m left, 0.5 m low, tilted 10°.



controller	max$\lvert\lambda(A-BK)\rvert$	result	max torque
slow	0.92	stable, still 0.13 m off after 6 s	0.010 N m
medium	0.80	stable, settled in about 3 s	0.049 N m
fast	0.55	stable, settled in about 1.3 s, big tilt swings	0.43 N m
3 × medium	**1.66**	**unstable**	explodes

What to notice

- The grey crosses (open loop) sit at 1. **Feedback pulls them into the circle**, and the further in, the faster the drone recovers. This is Experiment 1 again: the largest $\lvert\lambda\rvert$ sets the settling speed.
- **Fast costs effort:** the fast controller uses about 9× more torque than the medium one and tilts the drone more than 20°, which is beyond what the small-angle model is good for.

- **More gain is not always better.** Tripling the medium gain puts an eigenvalue at $\lambda = -1.66$, outside the circle. Because it is *negative*, the drone over-corrects and flips direction at every step, with growing amplitude. In discrete time, the controller only acts every T_s, so too strong a correction overshoots.

Animation "Feedback recovery": open `lab/figures/exp5_feedback_recovery.gif` in the repository.

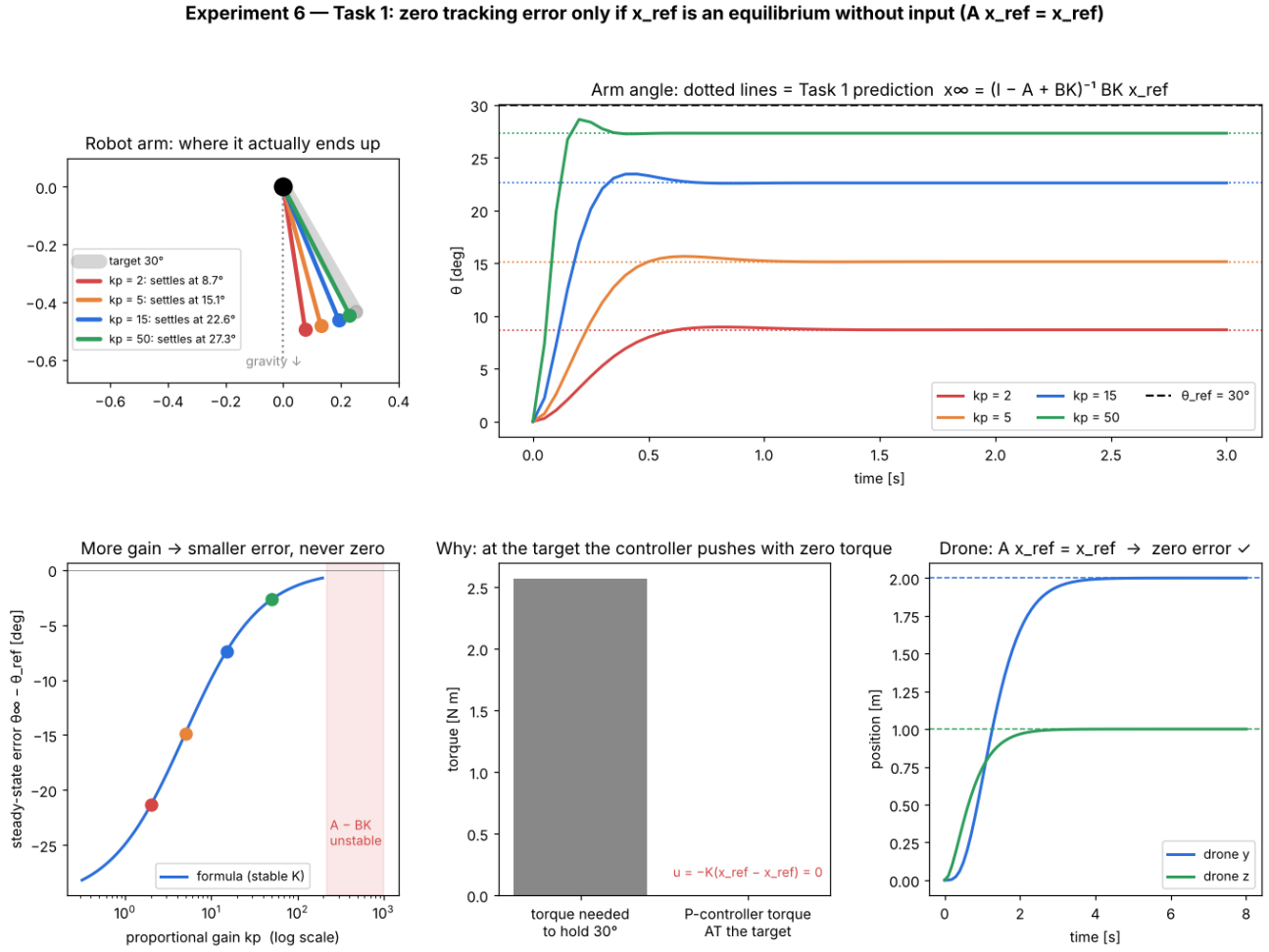
Try it yourself: change the factor `3 * K_med` to `2 * K_med`. Is it still stable? Find, by trial, the largest factor that keeps every $|\lambda| < 1$. (Answer: 2× is still stable with max $|\lambda| = 0.86$; the limit lies between 2× and 2.5×.)

Experiment 6 — Task 1 live: the tracking error

Script: `lab/exp6_tracking_error.py` · **Concepts:** Concepts §7.2–7.3 · Solutions Task 1

Now the controller tracks a reference, $u_k = -K(x_k - x_{\text{ref}})$. Task 1 showed that

$$x_\infty - x_{\text{ref}} = (I - A + BK)^{-1}(A - I)\, x_{\text{ref}}, \qquad \text{which is zero} \iff Ax_{\text{ref}} = x_{\text{ref}}.$$



The drone: zero error (bottom-right)

The target is "hover 2 m right, 1 m up": $x_{\text{ref}} = [2, 1, 0, 0, 0, 0]$. A drone hovering at rest stays where it is with $u = 0$, because positions are integrators. So $Ax_{\text{ref}} = x_{\text{ref}}$:

```
DRONE: A x_ref == x_ref ? True
  predicted error (I-A+BK)^-1 (A-I) x_ref = [0. 0. 0. 0. 0. 0.]
  simulated final state                   = [2. 1. 0. 0. 0. 0.]
```

The drone lands exactly on the target. This is the "condition holds" case of Task 1.

The robot arm: steady-state error (everything else)

The target is "hold the arm at 30°": $x_{\text{ref}} = [30°, 0]$. Gravity pulls the arm back, so after one step from rest at 30° it would not stay there:

```
ARM: A x_ref = [0.5109 -0.5019]  vs  x_ref = [0.5236 0.]  -> equal? False
```

The condition fails, so **no gain** gives zero error:

k_p	settles at (formula)	settles at (simulation)	error
2	8.69°	8.69°	−21.3°
5	15.14°	15.14°	−14.9°
15	22.61°	22.61°	−7.4°
50	27.32°	27.32°	−2.7°

The simulated values match the Task 1 formula to every digit shown.

Why it happens (bottom-middle panel). To hold 30°, the motor must supply the holding torque $mgl\,\theta_{\text{ref}} \approx 2.6$ N m. But at the target the P-controller outputs $u = -K(x_{\text{ref}} - x_{\text{ref}}) = 0$. The arm therefore settles where the remaining error produces exactly the torque needed: $k_p(\theta_{\text{ref}} - \theta_\infty) = mgl\,\theta_\infty$. Solving gives

$$\theta_\infty = \frac{k_p}{k_p + mgl}\,\theta_{\text{ref}},$$

which is the Task 1 result $x_\infty = (I - A + BK)^{-1}BKx_{\text{ref}}$ written in physical terms for the arm. With $mgl = 4.9$ N m: $k_p = 5$ gives $\theta_\infty \approx 0.505 \cdot 30° = 15.1°$ ✓.

More gain helps, but it can't fix the problem (bottom-left). The error shrinks as k_p grows but never reaches zero. Beyond $k_p \approx 216$ the discrete closed loop becomes unstable (red band), just like the gain limit in the 1-D example of Task 1.2 and in Experiment 5.

Animation "Arm sagging": open `lab/figures/exp6_arm_sag.gif` in the repository.

Try it yourself

- Change the damping ratio `zeta` in `arm_gain`. The transient changes, but the final angle does not. (k_d multiplies ω, and $\omega_{\text{ref}} = 0$, so it never enters the error.)
- Set `theta_ref = 0`. The error vanishes, because $x_{\text{ref}} = 0$ trivially satisfies $Ax_{\text{ref}} = x_{\text{ref}}$.

This sag is what **Task 2** (a reference prefilter F that aims higher on purpose) and **Task 3** (an integrator that keeps adding torque until the error is zero) are designed to remove.

Summary: what each formula looks like in real life

Formula / term	In the lab
λ with $\lvert\lambda\rvert < 1$	a disturbance that dies out (Exp. 1, 5)
$\lambda = 1$	something that stays where you leave it: a hovering drone's position (Exp. 2)
repeated $\lambda = 1$, too few eigenvectors	a tilt that makes the drone drift faster and faster (Exp. 2)
rank $W_c = n$	the motors can put the drone anywhere (Exp. 3)
reachable subspace	the only places a broken drone can still go (Exp. 3)
minimum number of steps	the first N where the rank of $W_c(N)$ is full; feasible, but with huge inputs (Exp. 3)
$\ker W_o$	a drift the altimeter cannot see (Exp. 4)
rank $W_o = n$	the sensors are enough to reconstruct the full state (Exp. 4)
$A - BK$	the drone *with* its controller; its eigenvalues decide speed and stability (Exp. 5)
$(I - A + BK)^{-1}(A - I)x_{\text{ref}}$	how far the arm sags below the target (Exp. 6)
$Ax_{\text{ref}} = x_{\text{ref}}$	"the target is a place where the system would stay without any input" (Exp. 6)